MAKING A FINITE MORPHISM OVER AN AFFINE BASE LOCALLY FREE BY A FINITE LOCALLY FREE BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Splitting of finitely many monic polynomials by a finite locally free ring extension. Algebra, Lemma \ref{algebra-lemma-adjoin-roots}, cited at proof lines 12471-12478. Produces the finite locally free base change.
- Flatness descent along a finite locally free base change. More on Algebra, Section \ref{more-algebra-section-descent-flatness-integral}, cited at proof lines 12467-12469. Frames the flatness statements needed for the locally free assertions.

Source reference. Original lemma 03HW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4582–4582. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 052Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6586–6586. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 03HS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4578–4578. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $f : Y \to X$ be a finite morphism with X affine. There exists a diagram*

$$\begin{array}{ccccc} Z' & \xleftarrow[i]{} & Y' & \longrightarrow & Y \\ & \searrow & \downarrow & & \downarrow \\ & & X' & \longrightarrow & X \end{array}$$

where

(1) *$Y' \to Y$ and $X' \to X$ are surjective finite locally free,*
(2) *$Y' = X' \times_X Y$,*
(3) *$i : Y' \to Z'$ is a closed immersion,*
(4) *$Z' \to X'$ is finite locally free, and*
(5) *$Z' = \bigcup_{j=1,\ldots,m} Z'_j$ is a (set theoretic) finite union of closed subschemes, each of which maps isomorphically to X'.*

Proof. Write $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$. See also More on Algebra, Section 052Y. Let $x_1, \ldots, x_n \in B$ be generators of B over A. For each i we can choose a monic polynomial $P_i(T) \in A[T]$ such that $P(x_i) = 0$ in B. By Algebra, Lemma 03HS (applied n times) there exists a finite locally free ring extension $A \subset A'$ such that each P_i splits completely:

$$P_i(T) = \prod\nolimits_{k=1,\ldots,d_i} (T - \alpha_{ik})$$

for certain $\alpha_{ik} \in A'$. Set

$$C = A'[T_1, \ldots, T_n]/(P_1(T_1), \ldots, P_n(T_n))$$

and $B' = A' \otimes_A B$. The map $C \to B'$, $T_i \mapsto 1 \otimes x_i$ is an A'-algebra surjection. Setting $X' = \mathrm{Spec}(A')$, $Y' = \mathrm{Spec}(B')$ and $Z' = \mathrm{Spec}(C)$ we see that (1) – (4) hold. Part (5) holds because set theoretically $\mathrm{Spec}(C)$ is the union of the closed subschemes cut out by the ideals

$$(T_1 - \alpha_{1k_1}, T_2 - \alpha_{2k_2}, \ldots, T_n - \alpha_{nk_n})$$

for any $1 \leq k_i \leq d_i$. □